

MD IRSHAD

Lead AI Platform & MLOps Engineer

+91-9097421696 | abdusirshad@gmail.com | linkedin.com/in/md-irshad-devops | github.com/abdusirshad | abdusirshad.github.io/portfolio | Delhi, India

SUMMARY

- Lead AI Platform & MLOps Engineer and **Platform Engineering** practitioner with **7+ years** shipping production infrastructure on **Azure** and **AWS** for regulated banking, fintech, and enterprise SaaS – including two AI platforms (**TRecos**, **NUMO SDLC**) delivered to **First Abu Dhabi Bank (FAB), UAE**.
- Holding **CKA** (Linux Foundation) and **AZ-104** (Microsoft Azure) credentials; architected **end-to-end RAG pipelines** (Azure OpenAI gpt-4o + text-embedding-3-small, pgvector 1536-dim), multi-store data layers (PostgreSQL, Cosmos DB, Blob Storage), and **GitOps**-driven deployments with **Argo CD** and **Argo Workflows**.
- Built and owned **DevSecOps**-hardened CI/CD pipelines – **Trivy** container scanning (zero HIGH/CRITICAL gate), **SonarQube** SAST, **Snyk** SCA, **OWASP ZAP** DAST – with secrets secured via **HashiCorp Vault**, **Azure Key Vault**, and **AWS KMS**; upheld **SRE** standards including **SLO/SLI** monitoring and incident response.
- Administered **750+ Linux servers** at **99.99% availability**; achieved **25–40% cost savings** via **Karpenter** spot-instance auto-scaling and **FinOps** lifecycle policies; spearheaded **AWS cloud migration** cutting provisioning from days to hours.
- Delivered end-to-end **Observability** stacks (Azure Monitor, Prometheus, Grafana, ELK, Langfuse) and **IaC**-driven platform provisioning with **Terraform** and **Helm** across multi-tenant, multi-region cloud environments.

TECHNICAL SKILLS

Microsoft Azure: AKS, Azure OpenAI (gpt-4o, text-embedding-3-small), Cosmos DB, Blob Storage, ACR, App Gateway, Key Vault, Azure AD / Entra ID, Azure DevOps, VNet, NSG, Private Endpoints, Azure Monitor, Load Balancer, VM, Function Apps
Amazon Web Services (AWS): EKS, ECS, EC2, RDS, S3, IAM, Lambda, CloudFormation, SSM, KMS, Route 53, Auto Scaling

AI & MLOps: Azure OpenAI Service, RAG Pipelines, pgvector, Langfuse, Argo Workflows, LangChain, Amazon Bedrock
Containers & Orchestration: Docker, Kubernetes (EKS, AKS), Helm, Karpenter, Istio, Ingress, ConfigMaps, NVIDIA GPU Node Pools

CI/CD & GitOps: Azure DevOps, Jenkins, GitHub Actions, GitLab CI, Argo CD, Argo Workflows, Harness

Infrastructure as Code (IaC): Terraform, CloudFormation, Ansible, Helm

Security & DevSecOps: Trivy, SonarQube (SAST), Snyk (SCA), OWASP ZAP (DAST), HashiCorp Vault, Azure Key Vault, AWS KMS, IAM, MSAL JWT, Microsoft Graph API

Observability & SRE: Prometheus, Grafana, Azure Monitor, OpenTelemetry, Langfuse, CloudWatch, ELK Stack, SLO/SLI, Incident Management

FinOps & Cost Governance: Karpenter Spot Scaling, EC2 Scheduling, Snapshot Lifecycle Policies, Cost Attribution

Scripting & Frameworks: Python 3.12, Bash, FastAPI, PowerShell, Groovy

Databases: PostgreSQL + pgvector, Azure Cosmos DB, Amazon RDS, DynamoDB, Aurora, MySQL, MongoDB

EXPERIENCE

SiriusAI (*Cloud-Native Enterprise AI Platform Company*)

Oct 2025 – Present

Lead AI Platform & MLOps Engineer | Team of 4 Engineers

Gurugram, India

- Designed and deployed **TRecos** real-time recommendation engine on **AKS** with **NVIDIA GPU node pools**, enabling sub-second banking-grade recommendations – **Argo Workflows** for ML training orchestration, **Azure OpenAI** (gpt-4o + text-embedding-3-small) for inference, and **pgvector** RAG pipeline with 1536-dim cosine similarity retrieval; reduced model-to-production cycle from weeks to under 3 days.
- Architected and shipped **NUMO SDLC** AI delivery platform for **First Abu Dhabi Bank (FAB, UAE)** – orchestrating BRD authoring, Jira story generation, Confluence page creation, and BDD feature file commits via **FastAPI** + **Azure OpenAI** RAG pipeline on **AKS**; reduced analyst-to-delivery cycle from **3–5 days to under 4 hours**.
- Enforced **Azure AD** group-based **RBAC** across **6 NUMO product modules** using **MSAL JWT** validation and **Microsoft Graph API** for group overage resolution; integrated Jira and Confluence via **OAuth 2.0**, enabling secure **multi-tenant** access for **2 enterprise user groups** without custom identity infrastructure.
- Built **multi-store data architecture** supporting concurrent BRD sessions: **PostgreSQL** + **pgvector** (RAG embeddings), **Azure Cosmos DB** (TTL-based multi-turn chat history, session_id partition key), and **Azure Blob Storage** (document persistence); instrumented with **Langfuse** achieving full LLM trace and cost attribution across all modules.
- Owned end-to-end **DevSecOps CI/CD pipelines** for both platforms – docker build –> **Trivy** scan (HIGH/CRITICAL gate, exit-code 1) –> **ACR** push –> helm upgrade on **AKS** – reducing deployment lead time by **70%** with **zero HIGH/CRITICAL** vulnerabilities reaching production.
- Provisioned and scaled **AKS cluster infrastructure** via **Terraform** and **Helm** – dual namespace isolation (solutions-sdlc, solutions-trecos), App Gateway ingress with TLS termination, ACR managed identity, and **Azure Monitor** dashboards; achieved **99.9% pod availability** across both platforms.

- Led **Azure tenant migration** (India to US region) using **Terraform**, re-architecting VNets, Entra ID, Key Vault, AKS clusters, and ACR with **zero data loss** and **zero downtime**; completed full rebuild of CI/CD pipelines with **SRE-aligned** policy gates within **4 weeks**.

Biz2Credit (*SaaS-based Global Fintech Company*)

Oct 2021 – Sep 2025

Senior DevOps Engineer | Team of 6 Engineers

New Delhi, India

- Engineered highly available **AWS architectures** (EC2, RDS, S3, Route 53) across **3 AWS regions**, sustaining **99.99% SLA** for critical fintech applications processing **10,000+ daily loan transactions** for global customers.
- Streamlined infrastructure provisioning using **Terraform** and **CloudFormation** (IaC), reducing manual setup time **by 90%** and delivering consistent, repeatable environments across dev, staging, and production for **30+ microservices**.
- Built and oversaw **GitOps-driven CI/CD pipelines** with **Jenkins** and **GitLab CI**, accelerating software delivery **by 40%** through parallel jobs and policy-enforced quality gates across **10+ application teams**.
- Administered **750+ Linux servers** across AWS, orchestrating patching, performance tuning, and monitoring via **Ansible** and **Shell scripts** to uphold **99.99% availability** and cut operational toil **by 60%**.
- Deployed and operated **production-grade EKS clusters** hosting **50+ microservices**, with reusable Helm charts, ingress controllers, and zero-downtime rolling upgrades across all environments.
- Introduced **Karpenter-based FinOps** spot-instance auto-scaling on EKS, cutting **compute costs by 30–40%** and applying S3 snapshot lifecycle policies to reduce storage costs **by 25%** – delivering combined infrastructure savings of **\$200k+ annually**.
- Developed **Python** and **Bash** scripts to systematize EC2 lifecycle management, log rotation, and backup scheduling, improving operational efficiency **by 50%** and eliminating **20+ hours/week** of manual operational overhead.
- Established **multi-layered AWS backup strategies** enabling **60-day point-in-time recovery** for **15+ production RDS databases** with **SLO-aligned** scheduled recovery testing and audit-ready compliance reporting.
- Spearheaded **migration of 40+ services to AWS**, cutting deployment time **by 70%** and infrastructure provisioning from **3 days to 2 hours**; onboarded **10+ engineering teams**, reducing release-related incidents **by 30%**.
- Operated **HA MySQL and MongoDB** clusters with replication and automated failover; optimized Docker image builds achieving **20–30%** faster container start times across all services.
- Hardened security via **AWS IAM / KMS / Systems Manager** and certificate rotation, mitigating **100+ vulnerabilities**; directed incident response workflows and drove **MTTR reduction by 40%** across platform teams.

Tata Consultancy Services (*Global IT Service Company*)

Feb 2019 – Sep 2021

DevOps Engineer

Gurugram, India

- Led the first deployment of **BaNCs core banking solution** on **AWS** for a major financial client, establishing cloud-native banking infrastructure in a regulated environment.
- Architected secure **AWS infrastructure** (EC2 AMI-based, VPC, Security Groups, Route 53, EFS, CloudWatch) for high availability and centralized **observability**.
- Built **CI/CD pipelines** with **Jenkins** and **GitLab CI** enabling continuous delivery; integrated with code repositories and test automation frameworks to improve developer productivity.
- Created custom **Bash scripts** and cron jobs to systematize operational tasks, reducing **manual effort by 50%** and improving platform reliability.
- Set up load balancing and reverse proxy infrastructure for traffic routing and horizontal scaling of banking application tiers.
- Configured monitoring and alerting with **AWS CloudWatch**, establishing **SLI-based** visibility for proactive issue detection and operational response.

PROJECTS

TRecos – Real-Time AI Recommendation Engine

github.com/abdusirshad

AKS | NVIDIA GPU | Argo Workflows | Azure OpenAI | pgvector | Terraform

SiriusAI, 2025 – Present

- Built end-to-end ML platform on **AKS with NVIDIA GPU node pools** – **Argo Workflows** for training pipeline orchestration, **gpt-4o + text-embedding-3-small** for real-time inference, and **pgvector** 1536-dim cosine similarity RAG for banking-grade personalized recommendations.
- Provisioned all infrastructure via **Terraform** and **Helm**; applied **DevSecOps CI/CD** (Trivy zero-vulnerability gate, ACR, Azure DevOps) achieving **99.9% pod availability** and sub-second recommendation latency in production.

NUMO SDLC – AI-Driven Software Delivery Platform

github.com/abdusirshad

FastAPI | React 18 | AKS | Azure OpenAI | pgvector | Cosmos DB | Langfuse

SiriusAI / FAB UAE, 2025 – Present

- Architected a **multi-tenant AI SDLC automation platform** for **First Abu Dhabi Bank** – 6 RBAC-gated modules (BRD, Jira, Confluence, Design, Pair Programming, Testing) secured via **Azure AD MSAL JWT** and **Microsoft Graph API**; reduced analyst-to-delivery cycle **from 3–5 days to under 4 hours**.
- Built **RAG pipeline** (FastAPI + pgvector + Azure OpenAI) with **SSE streaming**, **Cosmos DB** multi-turn history, **Jira / Confluence OAuth 2.0** integration, and **Langfuse** LLM tracing; deployed on **AKS** with full **GitOps CI/CD** and **OpenTelemetry** observability.

CERTIFICATION

Certified Kubernetes Administrator (CKA) – Linux Foundation	2025
Microsoft Certified: Azure Administrator Associate (AZ-104)	2025
Microsoft Certified: Azure Fundamentals (AZ-900)	2025
AWS Certified Solutions Architect – Associate (SAA-C03)	Scheduled: Sep 2026 (In Progress)

EDUCATION

Muzaffarpur Institute of Technology (MIT)	Jul 2014 – Aug 2018
Bachelor of Technology (B.Tech) in Information Technology (IT)	Muzaffarpur, Bihar